Minh Manh Tuan Nguyen

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EDUCATION

University of Georgia Expected May 2028
Bachelor of Science in Computer Systems Engineering, Honors College GPA: 3.81

WORK EXPERIENCE

Greenhouse Robotics Lab | Athens, GA Sep 2026 – Present
Undergraduate Researcher

- Lead end-to-end development of a rail-mounted manipulator for autonomous greenhouse plant monitoring, owning requirements, architecture, a $2.5K+ bill of materials, and a scaled proof-of-concept testbed.
- Design an overhead dual-drive XY gantry for an inverted 6-DOF arm and integrate G-code motion firmware with a ROS 2 hardware interface for repeatable sub-millimeter positioning.
- Develop a coarse-to-fine localization pipeline fusing overhead AprilTag tracking with a wrist-mounted RealSense depth camera, using hand-eye calibration and visual servoing to place and retrieve sensors on plant leaves.
- Architect an MQTT-based IoT bridge and task state machine that trigger autonomous place, retrieve, and verify cycles from sensor telemetry or operators, with a hardware e-stop and software interlocks.

Disassembly Robotics Lab | Athens, GA May 2026 – Present
Undergraduate Researcher

- Design and implement ROS2 action-server architecture for collision-aware UR5e motion planning, integrating ROS2 libraries to replace classical control logic with planning-scene-based collision avoidance.
- Architect inter-node communication contracts between perception, state-machine orchestration, and robot control nodes for an automated gearbox disassembly pipeline.

UGA Small Satellite Research Laboratory | Athens, GA Dec 2025 – Present
MOCI Chief Engineer

- Direct the C-based flight-software architecture for five subsystems, using Docker containers to build a repeatable development pipeline that improves reliability and reduces integration errors.
- Develop C90 flight-software scripts with the Bright Ascension framework to automate spacecraft state-machine transitions, control loops, and power sequencing, enabling reliable autonomous operation.
- Debug embedded anomalies during hardware-in-the-loop testing using oscilloscopes and encoder diagnostics, isolate non-volatile flash memory corruption, and restore all flight modes.

Venuehorn | Remote, MA May 2026 – Aug 2026
ML/AI Engineering Intern

- Architected a chatbot and intent-extraction engine with FastAPI and OpenAI that converted natural-language queries into safe SQL, preventing injection attacks and enabling users to retrieve data instantly.
- Built an asynchronous computer-vision pipeline with Celery, OpenCV, and YOLOv8 that processed images from AWS S3 and stored extracted metadata in PostgreSQL, automating image analysis and tagging.

PROJECTS

Smart Biometric Polygraph | *ESP32-S3, HIL, Signal Processing* Aug 2026 – Present

- Designing a custom mixed-signal PCB in Altium Designer for a wrist-worn IoT device, engineering ground plane separation to isolate Wi-Fi noise from highly sensitive analog instrumentation amplifiers.
- Developing C/C++ ESP32-S3 firmware to interface with I2C sensors and applying Kalman filtering to raw optical PPG data, successfully extracting respiratory markers.

Self-Balancing Robot | *C++, MCU, Altium* May 2026 – Aug 2026

- Architected a cascaded PID control system on an ESP32 with anti-windup constraints, processing 100Hz IMU telemetry and encoder data to successfully maintain dynamic equilibrium and prevent motor stalling.
- Engineered a Hardware-in-the-Loop (HIL) tuning pipeline over Serial communication, enabling rapid iteration of physical control loops and real-time validation of balancing algorithms.

TECHNICAL SKILLS

Languages: C, C++, Python, Javascript, Verilog
Tools & Frameworks: Linux, ROS2, OpenCV, Docker, Git, CI/CD, Altium Designer, Matlab, Simulink
Hardware: Raspi, MCU, Communication Protocols, Oscilloscopes, Logic Analyzers, PCB Design, FPGAs